

G8:

Mounted at /content/drive

DATA_DIR: /content/drive/MyDrive/convergence_experiment | exists: True

OUT_DIR : /content/drive/MyDrive/convergence_experiment/G8_outputs

G9_fractal/g9_frac_cnn_9533.npz 78.2 MB

G9_fractal/g9_frac_cnn_inorm_9533.npz 78.2 MB

G9_fractal/g9_frac_vit_9533.npz 14.7 MB

G9_fractal/g9_nat_cnn_9533.npz 78.2 MB

G9_fractal/g9_nat_vit_9533.npz 14.7 MB

G9_fractal/g9_pixels_9533.npz 117.2 MB

G9_fractal/g9_rand_cnn_9533.npz 78.2 MB

G9_fractal/g9_rand_vit_9533.npz 14.7 MB

activations.npz 840.3 MB

adapter.npz 3.7 MB

adapter_fp16.npz 0.7 MB

cka_matrices.npz 0.0 MB

crossmodal_adapter.npz 6.0 MB

crossmodal_pairs.npz 134.6 MB

crossmodal_pairs_gpt2.npz 117.6 MB

crossmodal_pairs_gpt2_dinov2-large_cls+patch.npz 183.9 MB

distill_ckpt.npz 189.8 MB

distill_corpus.npz 190.8 MB

e11_img_ckpt_cls.npz 14.4 MB

e13_txt_bert.npz 51.3 MB

e13_txt_sbert.npz 51.4 MB
 e1_img_ckpt.npz 14.4 MB
 e1_img_ckpt_convnext-base-224-22k_native.npz 36.3 MB
 e1_img_ckpt_dinov2-base_cls+patch.npz 65.8 MB
 e1_img_ckpt_dinov2-large_cls+patch.npz 115.8 MB
 e1_img_ckpt_dinov2-small_cls+patch.npz 17.3 MB
 e1_img_ckpt_siglip2-base-patch16-224_native.npz 27.1 MB
 f5_eval_ckpt.npz 43.6 MB
 figs_for_claude/shape_agreement_matrix.npz 0.0 MB
 g7_rebuilt.npz 0.0 MB
 hub_exact.npz 227.4 MB
 hub_rebuilt.npz 831.3 MB
 pairs.npz 30.5 MB
 scale_extension/cache_img_giant.npz 70.0 MB
 scale_extension/cache_txt_qwen06.npz 28.9 MB
 scale_extension/cache_txt_qwen4.npz 72.0 MB
 scale_extension/rho_matrix_extended.npz 0.0 MB

,PASS mutual_knn(X,X)	got=1.0000 want~1.0000
PASS cka_linear(X,X)	got=1.0000 want~1.0000
PASS cka_rbf(X,X)	got=1.0000 want~1.0000
PASS relative_agreement(X,X)	got=1.0000 want~1.0000
PASS shape_agreement(X,X)	got=1.0000 want~1.0000
PASS mutual_knn rotation-invariant	got=1.0000 want~1.0000
PASS cka_linear rotation-invariant	got=1.0000 want~1.0000

PASS cka_linear isotropic-scale-invariant got=1.0000 want~1.0000

PASS relative_agreement rotation-invariant got=1.0000 want~1.0000

PASS shape_agreement rotation-invariant got=1.0000 want~1.0000

PASS mutual_knn independent $\sim k/(n-1)$ got=0.0307 want~0.0251

PASS cka_linear independent $\sim d/n$ (BIASED) got=0.1385 want~0.1600

PASS cka_linear_debiased independent ~ 0 got=0.0008 want~0.0000

PASS cka_linear_debiased(X,X) got=1.0000 want~1.0000

PASS cka_linear_debiased rotation-inv got=1.0000 want~1.0000

PASS relative_agreement independent ~ 0 got=0.0514 want~0.0000

PASS shape_agreement independent ~ 0 got=-0.0003 want~0.0000

noise=0.1 knn=0.809 cka=0.980 rel=0.980 rho=0.978

noise=0.5 knn=0.280 cka=0.637 rel=0.637 rho=0.616

noise=1.0 knn=0.083 cka=0.243 rel=0.261 rho=0.228

noise=2.0 knn=0.027 cka=0.041 rel=0.094 rho=0.045

PASS all four metrics decrease monotonically with noise

PASS shuffled mutual_knn $\sim$ chance got=0.0239 want~0.0251

PASS shuffled cka_linear $\sim d/n$ (BIASED) got=0.1379 want~0.1600

PASS shuffled cka_linear_debiased ~ 0 got=0.0000 want~0.0000

intact: knn=0.486 cka=0.861

PASS pca_whiten $\rightarrow$ identity covariance got=0.0000 want~0.0000

low-rank(5): alg=5 eff=4.95

anisotropic: alg=64 eff=1.06

PASS algebraic rank != effective rank on an anisotropic full-rank space

ALL TESTS PASSED

,g9_frac_cnn_9533.npz

emb (9533, 2048) float32

keep (9533,) int64

meta () <U204

g9_frac_cnn_inorm_9533.npz

emb (9533, 2048) float32

keep (9533,) int64

meta () <U216

g9_frac_vit_9533.npz

emb (9533, 384) float32

keep (9533,) int64

meta () <U301

g9_nat_cnn_9533.npz

emb (9533, 2048) float32

keep (9533,) int64

meta () <U180

g9_nat_vit_9533.npz

emb (9533, 384) float32

keep (9533,) int64

meta () <U214

g9_pixels_9533.npz

emb (9533, 3072) float32

keep (9533,) int64

meta () <U38

g9_rand_cnn_9533.npz

emb (9533, 2048) float32

keep (9533,) int64

meta () <U173

g9_rand_vit_9533.npz

emb (9533, 384) float32

keep (9533,) int64

meta () <U220

activations.npz

A_layers (13, 10000, 768) float32

B_layers (13, 10000, 768) float32

R_layers (13, 10000, 768) float32

n_samples () int64

seed () int64

adapter.npz

W_ridge (512, 768) float32

R_procrustes (768, 768) float32

proc_dims (2,) int64

train_idx (3000,) int64

eval_idx (1000,) int64

adapter_fp16.npz

W (512, 768) float16

cka_matrices.npz

trained (13, 13) float64

random (13, 13) float64

crossmodal_adapter.npz

W (768, 1024) float64

crossmodal_pairs.npz

img (18066, 1536) float32

txt (18066, 1024) float32

crossmodal_pairs_gpt2.npz

img (18066, 1536) float32

txt (18066, 768) float32

crossmodal_pairs_gpt2_dinov2-large_cls+patch.npz

img (23755, 2048) float32

txt (23755, 768) float32

distill_ckpt.npz

sig (39999, 768) float32

mob (39999, 512) float32

ids (39999,) int64

next () int64

teacher () <U31

distill_corpus.npz

sig (39999, 768) float32

mob (39999, 512) float32

ids (39999,) int64

eval_idx	(2000,) int64
train_idx	(37999,) int64
teacher	() <U31

e11_img_ckpt_cls.npz

img	(8000, 768) float32
keep	(8000,) int64
next	() int64

e13_txt_bert.npz

txt	(18066, 768) float32
-----	----------------------

e13_txt_sbert.npz

txt	(18066, 768) float32
-----	----------------------

e1_img_ckpt.npz

img	(8000, 768) float32
keep	(8000,) int64
next	() int64

e1_img_ckpt_convnext-base-224-22k_native.npz

img	(9533, 1024) float32
done	(1,) int64
keep	(9533,) int64

e1_img_ckpt_dinov2-base_cls+patch.npz

img	(18066, 1536) float32
keep	(18066,) int64
next	() int64
tag	() <U21

e1_img_ckpt_dinov2-large_cls+patch.npz

img (23755, 2048) float32

keep (23755,) int64

next () int64

tag () <U22

e1_img_ckpt_dinov2-small_cls+patch.npz

img (9533, 768) float32

keep (9533,) int64

next () int64

tag () <U22

e1_img_ckpt_siglip2-base-patch16-224_native.npz

img (9533, 768) float32

done (1,) int64

keep (9533,) int64

f5_eval_ckpt.npz

mob (5750, 512) float32

sig (5750, 768) float32

txt (5750, 768) float32

ids (5750,) int64

next () int64

shape_agreement_matrix.npz

names (8,) <U9

M (8, 8) float64

n_items () int64

g7_rebuilt.npz

pair_names	(21,) <U21
gain	(21,) float64
pair_cos	(21,) float64
preserve	(7,) float64
encoder_names	(7,) <U9
spearman	() float64

hub_exact.npz

sv	(5376,) float64
basis	(5376, 512) float64
mu	(5376,) float64
hub_train	(8533, 512) float64
scale_per_space	(4,) float64
space_names	(4,) <U9
space_dims	(4,) int64
train_idx	(8533,) int64
eval_idx	(1000,) int64
n_pairs	() int64
hub_dim	() int64
alpha	() float64
train_on	(1,) <U9
head	(512, 1024) float64
to_hub_img_small	(768, 512) float64
to_hub_img_base	(1536, 512) float64

to_hub_img_large	(2048, 512) float64
to_hub_txt_bge	(1024, 512) float64
space_img_small	(9533, 768) float64
space_img_base	(9533, 1536) float64
space_img_large	(9533, 2048) float64
space_txt_bge	(9533, 1024) float64

hub_rebuilt.npz

S	(7680,) float64
V	(7680, 768) float64
mu	(7680,) float64
scale	(768,) float64
H_train	(8533, 768) float64
H_all	(9533, 768) float64
H_per_encoder	(7, 9533, 768) float64
fit_r2	(7,) float64
scale_vs_published	() float64
shape_err	() float64
top_err	() float64
raw_img_base	(9533, 1536) float64
raw_img_large	(9533, 2048) float64
raw_img_small	(9533, 768) float64
raw_txt_bert	(9533, 768) float64
raw_txt_bge	(9533, 1024) float64
raw_txt_gpt2	(9533, 768) float64

raw_txt_sbert	(9533, 768) float64
encoder_names	(7,) <U9
hub_ids	(9533,) int64
construction	(1,) <U12
n_train	() int64
n_eval	() int64
map_img_base	(1536, 768) float64
map_img_large	(2048, 768) float64
map_img_small	(768, 768) float64
map_txt_bert	(768, 768) float64
map_txt_bge	(1024, 768) float64
map_txt_gpt2	(768, 768) float64
map_txt_sbert	(768, 768) float64

pairs.npz

mob_img	(4000, 512) float32
sig_img	(4000, 768) float32
sig_txt	(4000, 768) float32

cache_img_giant.npz

v	(9533, 3072) float32
keep	(9533,) int64
next	() int64
tag	() <U22
params_b	() float64

cache_txt_qwen06.npz

v (9533, 1024) float64

hf () <U25

params_b () float64

pool () <U9

cache_txt_qwen4.npz

v (9533, 2560) float64

hf () <U23

params_b () float64

pool () <U9

rho_matrix_extended.npz

names (11,) <U10

M (11, 11) float64

n_items () int64

,loaded 8 of 8 encoders (all truncated to 9533 rows)

img_small (9533, 768) image dinov2 e1_img_ckpt_dinov2-small_cls+patch.npz

img_base (9533, 1536) image dinov2 e1_img_ckpt_dinov2-base_cls+patch.npz

img_large (9533, 2048) image dinov2 e1_img_ckpt_dinov2-large_cls+patch.npz

convnext (9533, 1024) image convnext
e1_img_ckpt_convnext-base-224-22k_native.npz

bge (9533, 1024) text contrastive-text crossmodal_pairs.npz

gpt2 (9533, 768) text causal-lm crossmodal_pairs_gpt2.npz

bert (9533, 768) text masked-lm e13_txt_bert.npz

sbert (9533, 768) text contrastive-text e13_txt_sbert.npz

	pair	rho	rho_shuffled	ratio	aligned
	img_small~gpt2	0.1633	0.0040	41.2520	True
	img_base~gpt2	0.1615	0.0023	70.9420	True
	img_large~convnext	0.3886	0.0044	87.8709	True
	convnext~sbert	0.3361	0.0037	90.4006	True
	gpt2~bert	0.3710	-0.0040	91.7999	True
	img_small~bert	0.3763	-0.0034	109.7615	True
	gpt2~sbert	0.1936	0.0018	110.3417	True
	img_base~convnext	0.3924	0.0034	113.9286	True
	img_base~img_large	0.5996	0.0051	116.5852	True
	convnext~gpt2	0.1759	0.0014	122.1984	True
	img_large~gpt2	0.1751	0.0014	126.6742	True
	bge~sbert	0.7734	-0.0053	145.8978	True
	img_base~bert	0.3800	-0.0024	157.5952	True
	img_small~convnext	0.3960	0.0024	165.9313	True
	img_small~bge	0.3017	0.0018	167.4593	True
	img_small~img_large	0.5528	0.0033	168.0134	True
	bge~gpt2	0.2209	0.0013	168.9321	True
	convnext~bge	0.3124	0.0018	169.1733	True
	img_large~bge	0.3241	-0.0016	203.1139	True
	img_large~bert	0.4128	-0.0016	262.9404	True
	bge~bert	0.3648	0.0012	303.9081	True
	img_small~sbert	0.3264	0.0011	304.6760	True
	img_large~sbert	0.3389	-0.0010	345.0384	True

flagged degenerate: {'gpt2'}

criteria: $\text{eff_rank}/\text{ambient} < 0.05$ (low-rank) OR $\text{mean_pair_cos} > 0.95$ (anisotropy) OR in {'gpt2'} (documented). GPT-2's problem is anisotropy, which a centered-eff-rank rule alone can miss -- so both criteria are reported and combined.,

n=4000 rows, k=10, anchors=512, whitened to 256d

mutual-kNN chance floor = 0.0025

done raw mutual_knn

done raw cka_debiased

done raw cka_biased

done raw relative

done raw shape_rho

done whitened mutual_knn

done whitened cka_debiased

done whitened cka_biased

done whitened relative

done whitened shape_rho, === mutual_knn [raw] ===

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
img_small	1.000	0.583	0.487	0.440	0.235	0.100	0.202	0.249
img_base	0.583	1.000	0.634	0.456	0.250	0.103	0.206	0.263
img_large	0.487	0.634	1.000	0.430	0.255	0.104	0.211	0.270
convnext	0.440	0.456	0.430	1.000	0.270	0.106	0.220	0.285
bge	0.235	0.250	0.255	0.270	1.000	0.190	0.451	0.735
gpt2	0.100	0.103	0.104	0.106	0.190	1.000	0.202	0.186
bert	0.202	0.206	0.211	0.220	0.451	0.202	1.000	0.454
sbert	0.249	0.263	0.270	0.285	0.735	0.186	0.454	1.000

=== cka_debiased [raw] ===

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
img_small	1.000	0.795	0.750	0.704	0.592	0.197	0.510	0.613
img_base	0.795	1.000	0.842	0.728	0.601	0.190	0.495	0.618
img_large	0.750	0.842	1.000	0.718	0.611	0.193	0.502	0.629
convnext	0.704	0.728	0.718	1.000	0.610	0.218	0.515	0.623
bge	0.592	0.601	0.611	0.610	1.000	0.280	0.738	0.904
gpt2	0.197	0.190	0.193	0.218	0.280	1.000	0.352	0.281
bert	0.510	0.495	0.502	0.515	0.738	0.352	1.000	0.747
sbert	0.613	0.618	0.629	0.623	0.904	0.281	0.747	1.000

=== relative [raw] ===

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
img_small	1.000	0.777	0.732	0.665	0.499	0.183	0.460	0.538
img_base	0.777	1.000	0.809	0.689	0.509	0.178	0.452	0.547
img_large	0.732	0.809	1.000	0.676	0.522	0.183	0.464	0.559
convnext	0.665	0.689	0.676	1.000	0.530	0.194	0.448	0.571
bge	0.499	0.509	0.522	0.530	1.000	0.256	0.586	0.866
gpt2	0.183	0.178	0.183	0.194	0.256	1.000	0.347	0.256
bert	0.460	0.452	0.464	0.448	0.586	0.347	1.000	0.606
sbert	0.538	0.547	0.559	0.571	0.866	0.256	0.606	1.000

=== shape_rho [raw] ===

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
img_small	1.000	0.591	0.561	0.401	0.287	0.159	0.380	0.315
img_base	0.591	1.000	0.603	0.398	0.283	0.158	0.376	0.304
img_large	0.561	0.603	1.000	0.398	0.317	0.168	0.413	0.330
convnext	0.401	0.398	0.398	1.000	0.307	0.167	0.303	0.335
bge	0.287	0.283	0.317	0.307	1.000	0.205	0.355	0.771
gpt2	0.159	0.158	0.168	0.167	0.205	1.000	0.346	0.194
bert	0.380	0.376	0.413	0.303	0.355	0.346	1.000	0.378
sbert	0.315	0.304	0.330	0.335	0.771	0.194	0.378	1.000

=== mutual_knn [whitened] ===

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
img_small	1.000	0.522	0.435	0.393	0.213	0.198	0.205	0.226
img_base	0.522	1.000	0.592	0.416	0.235	0.217	0.225	0.249
img_large	0.435	0.592	1.000	0.394	0.244	0.222	0.232	0.258
convnext	0.393	0.416	0.394	1.000	0.248	0.221	0.231	0.261
bge	0.213	0.235	0.244	0.248	1.000	0.456	0.473	0.630
gpt2	0.198	0.217	0.222	0.221	0.456	1.000	0.476	0.473
bert	0.205	0.225	0.232	0.231	0.473	0.476	1.000	0.491
sbert	0.226	0.249	0.258	0.261	0.630	0.473	0.491	1.000

=== cka_debiased [whitened] ===

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
img_small	1.000	0.424	0.358	0.325	0.187	0.163	0.163	0.196

img_base	0.424	1.000	0.472	0.350	0.221	0.189	0.189	0.231
img_large	0.358	0.472	1.000	0.342	0.237	0.202	0.201	0.246
convnext	0.325	0.350	0.342	1.000	0.228	0.195	0.194	0.240
bge	0.187	0.221	0.237	0.228	1.000	0.402	0.383	0.511
gpt2	0.163	0.189	0.202	0.195	0.402	1.000	0.384	0.392
bert	0.163	0.189	0.201	0.194	0.383	0.384	1.000	0.380
sbert	0.196	0.231	0.246	0.240	0.511	0.392	0.380	1.000

=== relative [whitened] ===

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
img_small	1.000	0.432	0.366	0.340	0.211	0.189	0.189	0.219
img_base	0.432	1.000	0.467	0.358	0.240	0.212	0.211	0.249
img_large	0.366	0.467	1.000	0.346	0.254	0.224	0.221	0.261
convnext	0.340	0.358	0.346	1.000	0.247	0.218	0.216	0.257
bge	0.211	0.240	0.254	0.247	1.000	0.420	0.402	0.523
gpt2	0.189	0.212	0.224	0.218	0.420	1.000	0.405	0.410
bert	0.189	0.211	0.221	0.216	0.402	0.405	1.000	0.398
sbert	0.219	0.249	0.261	0.257	0.523	0.410	0.398	1.000

=== shape_rho [whitened] ===

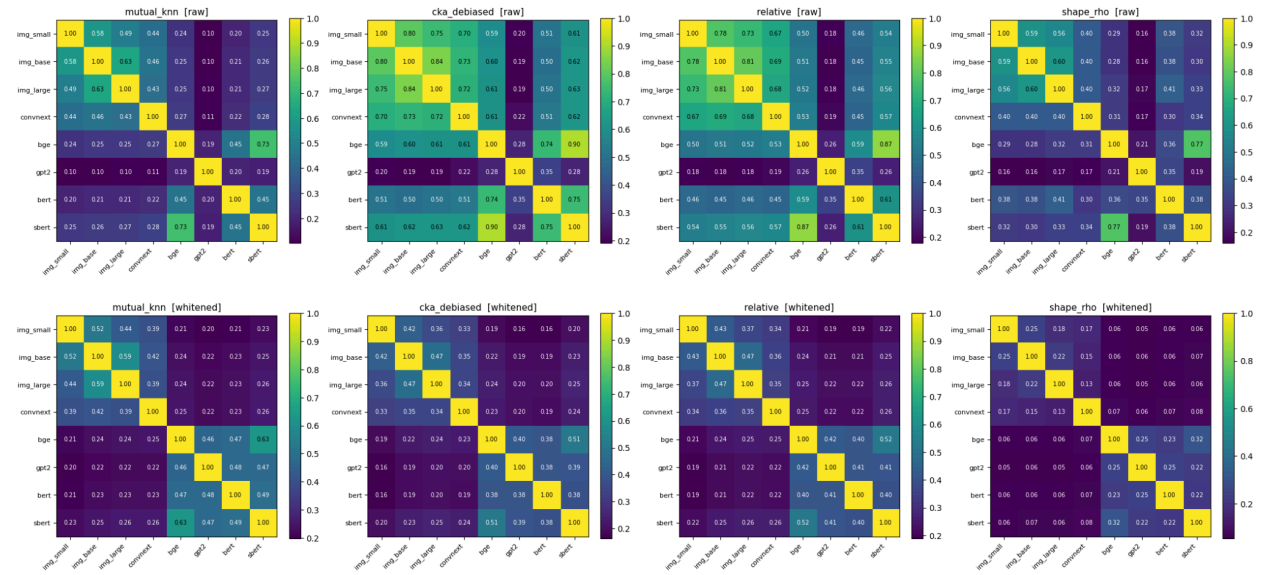
	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
img_small	1.000	0.246	0.176	0.167	0.060	0.054	0.058	0.063
img_base	0.246	1.000	0.222	0.154	0.064	0.055	0.059	0.066
img_large	0.176	0.222	1.000	0.131	0.063	0.054	0.058	0.065

convnext	0.167	0.154	0.131	1.000	0.075	0.063	0.067	0.076
bge	0.060	0.064	0.063	0.075	1.000	0.246	0.233	0.318
gpt2	0.054	0.055	0.054	0.063	0.246	1.000	0.249	0.223
bert	0.058	0.059	0.058	0.067	0.233	0.249	1.000	0.219
sbert	0.063	0.066	0.065	0.076	0.318	0.223	0.219	1.000

,

	intact	shuffled					
stat	cka_biased	cka_debiased	mutual_knn	relative	shape_rho	cka_biased	
	cka_debiased	mutual_knn	relative	shape_rho			
pair							
img_small~bge	0.5979	0.5921	0.2350	0.4987	0.2872	0.0188	0.0002
	0.0025	0.0162	-0.0003				
img_small~convnext	0.7149	0.7037	0.4404	0.6650	0.4009	0.0410	-0.0001
	0.0021	0.0338	0.0012				
img_small~img_base	0.8028	0.7955	0.5827	0.7772	0.5909	0.0376	-0.0001
	0.0025	0.0323	-0.0003				
img_small~img_large	0.7590	0.7498	0.4866	0.7324	0.5606	0.0385	-0.0001
	0.0023	0.0336	0.0004				

,



,

within-family (all) within-family (healthy-only) cross-family (all) cross-family
(healthy-only) cross-modal (all) cross-modal (healthy-only)

mutual_knn [raw] (0.6096, 4) (0.6096, 4) (0.3511, 8) (0.4461, 5)
(0.2079, 16) (0.2428, 12)

cka_debiased [raw] (0.8229, 4) (0.8229, 4) (0.5683, 8)
(0.7266, 5) (0.4824, 16) (0.5766, 12)

relative [raw] (0.7963, 4) (0.7963, 4) (0.5101, 8) (0.6444, 5)
(0.4273, 16) (0.5083, 12)

shape_rho [raw] (0.6314, 4) (0.6314, 4) (0.3343, 8) (0.386, 5)
(0.2878, 16) (0.3293, 12)

mutual_knn [whitened] (0.5448, 4) (0.5448, 4) (0.4465, 8)
(0.4333, 5) (0.2304, 16) (0.2357, 12)

cka_debiased [whitened] (0.4411, 4) (0.4411, 4) (0.3697, 8)
(0.356, 5) (0.2051, 16) (0.2111, 12)

relative [whitened] (0.4472, 4) (0.4472, 4) (0.3849, 8) (0.3688, 5)
(0.2263, 16) (0.2314, 12)

shape_rho [whitened] (0.2405, 4) (0.2405, 4) (0.2028, 8)
(0.1809, 5) (0.0624, 16) (0.0644, 12)

,source = img_small; targets = ['img_base', 'img_large', 'convnext'] (n = 3)

n < 4 -- report the ordering, not a correlation coefficient

view	stat	spearman	pearson	sim
raw	mutual_knn	-0.5	-0.002	[0.583, 0.487, 0.44]
raw	cka_debiased	-0.5	-0.203	[0.795, 0.75, 0.704]
raw	cka_biased	-0.5	-0.201	[0.803, 0.759, 0.715]
raw	relative	-0.5	-0.311	[0.777, 0.732, 0.665]
raw	shape_rho	-0.5	-0.544	[0.591, 0.561, 0.401]
whitened	mutual_knn	-0.5	-0.005	[0.522, 0.435, 0.393]
whitened	cka_debiased	-0.5	-0.004	[0.424, 0.358, 0.325]
whitened	cka_biased	-0.5	-0.001	[0.46, 0.398, 0.367]
whitened	relative	-0.5	0.037	[0.432, 0.366, 0.34]
whitened	shape_rho	-0.5	0.224	[0.246, 0.176, 0.167]

With four points every coefficient here is descriptive. The claim worth making is

ordinal: does the metric rank ConvNeXt above the DINOv2 targets, as transfer does?

,written to /content/drive/MyDrive/convergence_experiment/G8_outputs

G8_above_floor_heatmap.png

G8_clustered_heatmap.png

G8_config.json

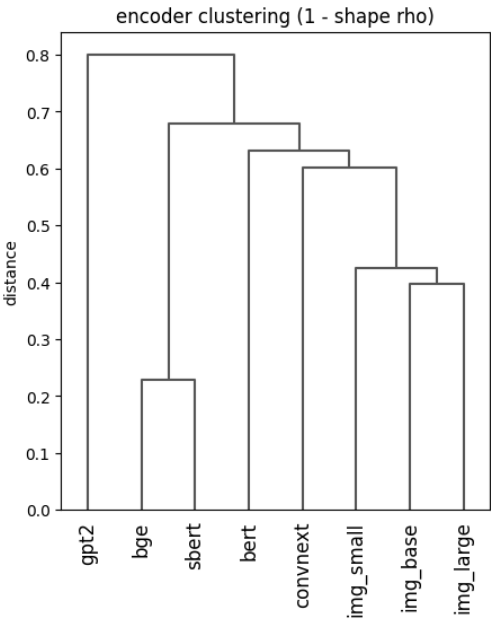
G8_matrices.xlsx

G8_metric_scatter.png

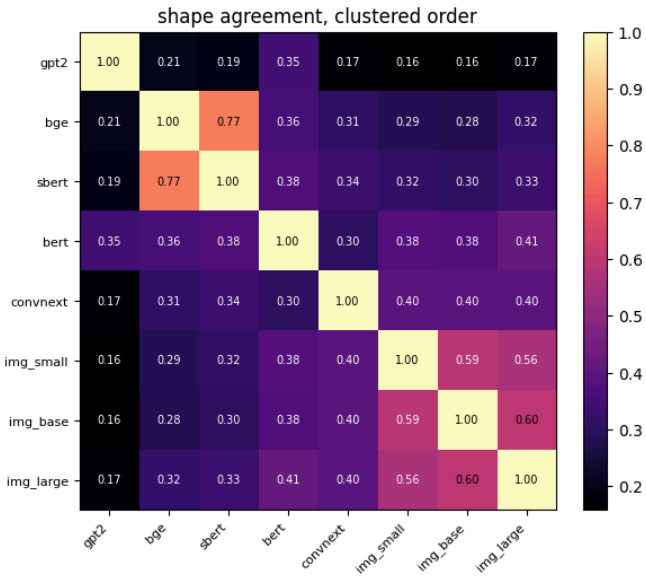
G8_similarity_matrices.png

G8_stratified_means.csv

,

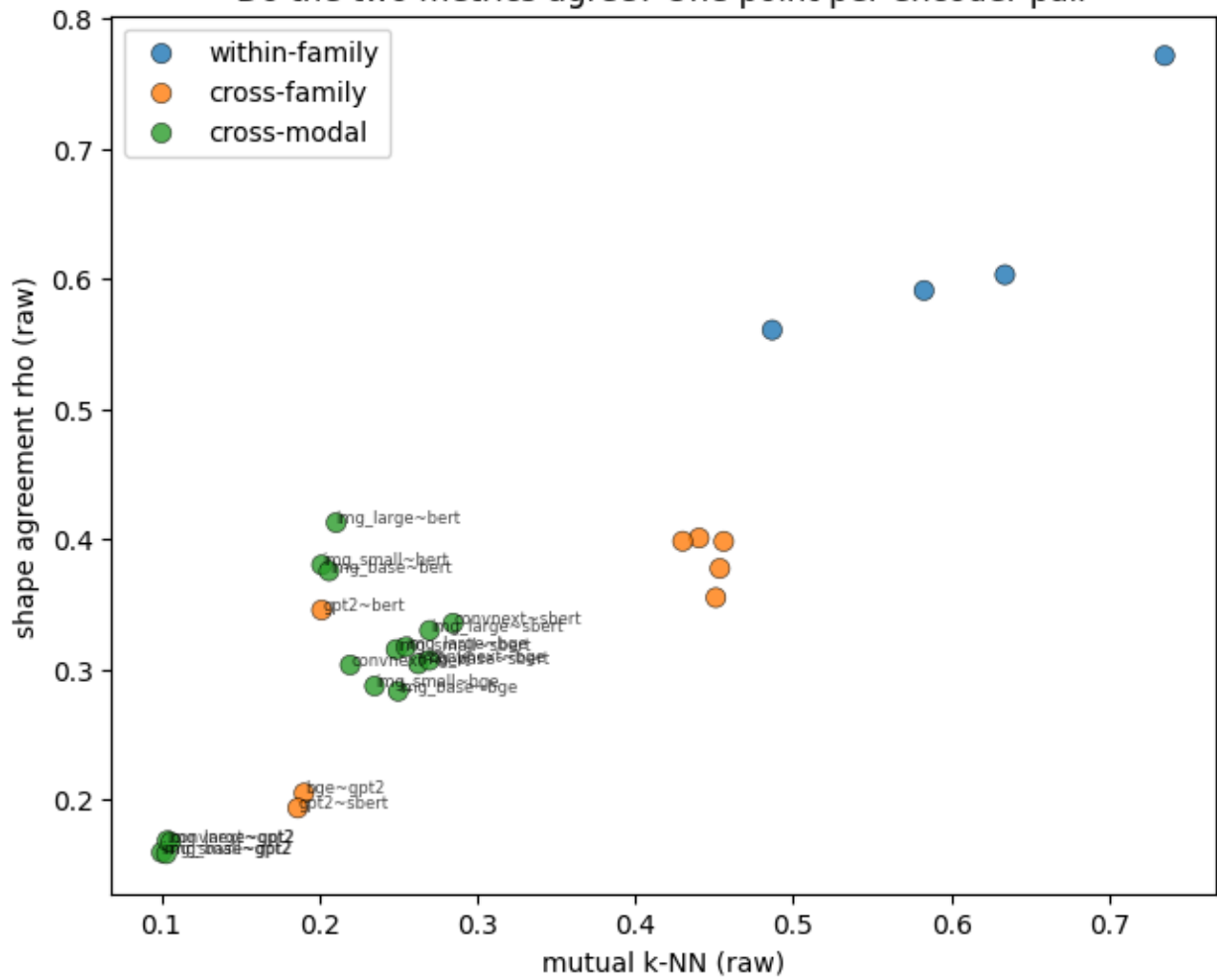


,

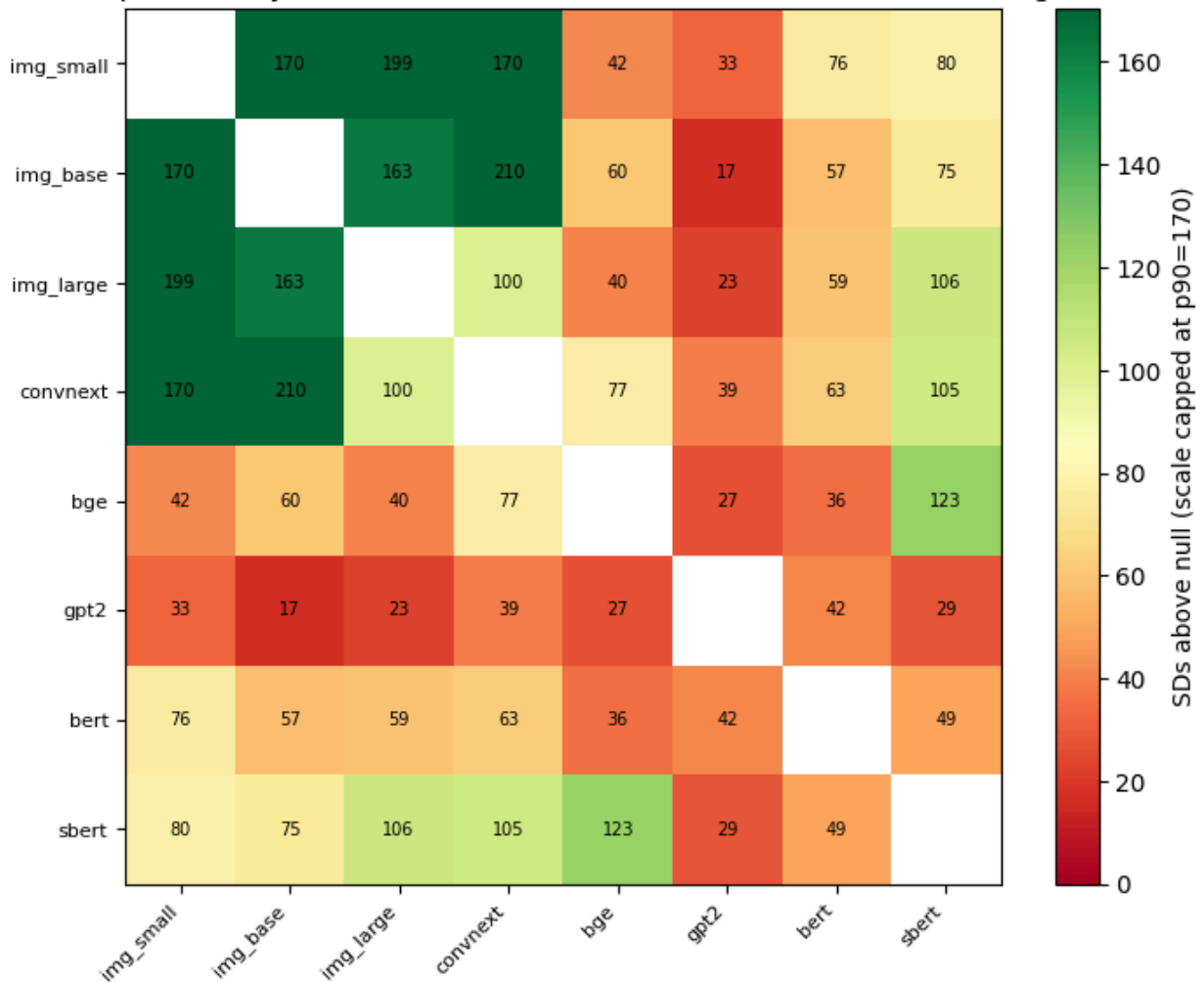


Scatter plot showing shape agreement ρ (raw) on the y-axis versus mutual k-NN (raw) on the x-axis. The plot is divided into three regions: within-family (blue), cross-family (orange), and cross-modal (green). Data points are labeled with model names.

Model Pair	mutual k-NN (raw)	shape agreement ρ (raw)	Category
img_large~bert	0.21	0.41	cross-modal
img_small~bert	0.20	0.38	cross-modal
img_base~bert	0.20	0.37	cross-modal
gpt2~bert	0.20	0.35	cross-family
img_large~sbert	0.27	0.33	cross-modal
img_small~sbert	0.26	0.32	cross-modal
img_base~sbert	0.25	0.31	cross-modal
convnext~bert	0.28	0.33	cross-modal
convnext~sbert	0.27	0.32	cross-modal
img_large~hoge	0.26	0.29	cross-modal
img_small~hoge	0.25	0.28	cross-modal
img_base~hoge	0.24	0.27	cross-modal
img_large~gpt2	0.19	0.20	cross-family
img_small~gpt2	0.18	0.19	cross-family
img_base~gpt2	0.17	0.18	cross-family
img_large~gpt2	0.11	0.16	cross-modal
img_small~gpt2	0.10	0.15	cross-modal
img_base~gpt2	0.09	0.14	cross-modal
gpt2~gpt2	0.73	0.77	within-family
img_large~img_large	0.63	0.60	within-family
img_small~img_small	0.58	0.59	within-family
img_base~img_base	0.48	0.56	within-family
gpt2~gpt2	0.45	0.40	cross-family
img_large~img_large	0.44	0.40	cross-family
img_small~img_small	0.43	0.40	cross-family
img_base~img_base	0.42	0.38	cross-family
gpt2~gpt2	0.45	0.36	cross-family



shape agreement above its shuffle floor (SDs, n_rep=20)
all pairs vastly > 0: this CONFIRMS realness, it is not a fine ranking



saved G8_clustered_heatmap.png, G8_metric_scatter.png, G8_above_floor_heatmap.png

note: SD-above-null confirms every pair is real; it does not rank pairs — use the shape_rho matrix itself for that.

G9:

timm 1.0.29 | gdown 5.2.2, Mounted at /content/drive

SYNTHETIC=False

DATA_DIR=/content/drive/MyDrive/convergence_experiment exists=True

WORK=/content/g9_work

OUT=/content/drive/MyDrive/convergence_experiment/G9_fractal

,PASS engine == scipy [shared] 0.711111 vs 0.711111

PASS engine == scipy [collapsed] 0.528727 vs 0.528727

PASS tie path == scipy average ranks

PASS rotation invariance

PASS shuffle null ~ 0 -0.0039

PASS R narrow bracket -> NaN

PASS R midpoint = 0.5

PASS RankBank == direct

ALL SELF-TESTS PASSED

					file key	rows_in_file	dim
,real run - skipped					encoder		
norm_published	norm_file	norm_prefix	identity_ok				
img_small	e1_img_ckpt_dinov2-small_cls+patch.npz	img	9533	768	54.70		
54.696	54.696	True					
img_base	e1_img_ckpt_dinov2-base_cls+patch.npz	img	18066	1536	52.91		
52.889	52.912	True					
img_large	e1_img_ckpt_dinov2-large_cls+patch.npz	img	23755	2048	50.83		
50.810	50.827	True					
convnext	e1_img_ckpt_convnext-base-224-22k_native.npz	img	9533	1024	16.70		
16.739	16.739	True					
bge	crossmodal_pairs.npz	txt	18066	1024	0.89	0.894	
0.894	True						
gpt2	crossmodal_pairs_gpt2.npz	txt	18066	768	207.78	208.211	
208.209	True						

bert	e13_txt_bert.npz txt	18066 768	9.15	9.183	9.186
True					
sbert	e13_txt_sbirt.npz txt	18066 768	2.43	2.427	2.428
True					

, ranked img_small ties=0.0e+00 (14s)

ranked img_base ties=0.0e+00 (28s)

ranked img_large ties=0.0e+00 (43s)

ranked convnext ties=0.0e+00 (57s)

ranked bge ties=2.2e-08 (71s)

ranked gpt2 ties=1.3e-06 (98s)

ranked bert ties=0.0e+00 (111s)

ranked sbirt ties=2.2e-08 (125s)

REPRODUCTION GATE (published E.12 / Table C12m values, tol 0.01)

	pair	published	measured	delta	ok
convnext~img_small	0.397	0.3971	0.0001	True	
convnext~img_base	0.392	0.3915	-0.0005	True	
convnext~img_large	0.391	0.3914	0.0004	True	
convnext~sbirt	0.330	0.3303	0.0003	True	
convnext~bert	0.305	0.3050	-0.0000	True	
convnext~bge	0.302	0.3025	0.0005	True	
convnext~gpt2	0.166	0.1662	0.0002	True	
bge~sbirt	0.768	0.7682	0.0002	True	

cross-modal mean over the 16 image-text pairs: 0.290 (published 8-encoder baseline 0.290)

VERIFIED -- the pipeline reproduces the report

,9533 COCO ids, 9 ... 46492 (published range 9 ... 46492),

1000/9533

2000/9533

3000/9533

4000/9533

5000/9533

6000/9533

7000/9533

8000/9533

9000/9533

download: {'train2017': 9533} (372s)

cropped 2000/9533

cropped 4000/9533

cropped 6000/9533

cropped 8000/9533

crops: (9533, 224, 224, 3) uint8 (385s total)

,checkpoints found:

w_vit/deitt16_224_fractal1k_lr3e-4_300ep.pth 69 MB

w_vit/deitt16_224_fractal10k_lr3e-4_100ep.pth 90 MB

w_resnet/FractalDB-10000_res18.pth 65 MB

w_resnet/FractalDB-10000_res152.pth 315 MB

w_resnet/FractalDB-10000_dense161.pth 195 MB

w_resnet/FractalDB-10000_res50.pth 176 MB

w_resnet/FractalDB-10000_bn_alexnet.pth 397 MB

w_resnet/FractalDB-10000_res34.pth 106 MB

w_resnet/FractalDB-10000_resnext101.pth 251 MB

w_resnet/FractalDB-1000_res50.pth 102 MB

w_resnet/FractalDB-1000_res18.pth 47 MB

w_resnet/FractalDB-1000_res34.pth 87 MB

w_resnet/FractalDB-10000_res101.pth 252 MB

w_resnet/FractalDB-1000_res152.pth 242 MB

w_resnet/FractalDB-1000_bn_alexnet.pth 250 MB

w_resnet/FractalDB-1000_res101.pth 179 MB

w_resnet/FractalDB-1000_resnext101.pth 177 MB

w_resnet/FractalDB-1000_dense161.pth 116 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_2.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_14.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_6.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_15.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_17.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_19.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_3.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_1.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_5.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_10.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_13.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_8.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_18.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_16.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_7.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_12.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_11.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_9.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_4.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_20.pth 102 MB

frac_cnn: FractalDB-1000_res50.pth sha256[:16]=12788a290bcd147e

frac_vit: deit16_224_fractal1k_lr3e-4_300ep.pth sha256[:16]=305227c6d58d570a

NOTE: the released fractal ViT is DeiT-TINY (~5M params) while the CNN is ResNet-50 (~25M). Each is bracketed against its OWN architecture's twins, so P1/P2/P3/P5 and R stay valid per architecture; only the CNN-vs-ViT contrast (X1) also carries a capacity gap -- flagged, not interpreted.

normalizations declared in FractalDB-Pretrained-ResNet-PyTorch:

score=1 pretraining/main.py: mean=(0.2, 0.2, 0.2) std=(0.5, 0.5, 0.5)

score=0 finetuning/main.py: mean=(0.485, 0.456, 0.406) std=(0.229, 0.224, 0.225)

-> using for frac_cnn: mean=(0.2, 0.2, 0.2) std=(0.5, 0.5, 0.5) [pretraining/main.py]

normalizations declared in FractalDB-Pretrained-ViT-PyTorch:

-> using for frac_vit: mean=(0.485, 0.456, 0.406) std=(0.229, 0.224, 0.225) [NOT FOUND in repo - ImageNet stats assumed]

,builders defined

, frac_cnn (9533, 2048) FractalDB-1000_res50.pth: 318 tensors loaded, 265 changed; []

frac_vit (9533, 384) deit16_224_fractal1k_lr3e-4_300ep.pth as
deit_tiny_patch16_224: 150 tensors loaded, 150 changed; ["unwrapped 'model'"]

rand_cnn (9533, 2048) random init, seed 0

rand_vit (9533, 384) random init deit_tiny_patch16_224, seed 0

nat_cnn (9533, 2048) torchvision IMAGENET1K_V1

nat_vit (9533, 384) timm deit_tiny_patch16_224.fb_in1k

pixels (9533, 3072) pixels 32x32 RGB, centred

frac_cnn_inorm (9533, 2048) FractalDB-1000_res50.pth: 318 tensors loaded, 265 changed; []

frac_vit_cls (9533, 192)

, space rho_vs_convnext shuffled ratio aligned

frac_cnn 0.0721 -0.0016 44.6 True

frac_vit 0.1243 0.0025 50.2 True

rand_cnn 0.0195 -0.0017 11.4 True

rand_vit 0.0675 0.0007 67.5 True

nat_cnn 0.3245 -0.0009 324.5 True

nat_vit 0.3473 -0.0012 294.1 True

pixels 0.0638 -0.0006 63.8 True

frac_cnn_inorm 0.0613 0.0023 26.4 True

frac_vit_cls 0.1246 0.0041 30.3 True

	space	dim	alg_rank	eff_rank	mean_pair_cos	mean_norm
frac_cnn	2048	2048	367.967		0.852	17.601
frac_vit	384	384	238.648		0.376	35.577
rand_cnn	2048	2034	256.140		0.998	1155.293
rand_vit	384	384	62.659		0.262	15.384
nat_cnn	2048	2048	909.757		0.569	29.297
nat_vit	384	384	278.472		0.123	27.024
pixels	3072	2999	966.584		-0.000	13.645
frac_cnn_inorm	2048	2048	315.729		0.919	15.407
frac_vit_cls	192	192	145.216		0.317	28.929
img_small	768	768	524.898		0.061	54.696
img_base	1536	1536	1008.660		0.049	52.912
img_large	2048	2048	1265.001		0.062	50.827
convnext	1024	1024	843.084		0.030	16.739
bge	1024	1024	424.653		0.569	0.894
gpt2	768	768	223.239		0.999	208.209
bert	768	768	395.370		0.807	9.186
sbert	768	766	368.722		0.136	2.428

collapsed by mean pair-cosine > 0.9: {'gpt2', 'rand_cnn', 'frac_cnn_inorm'} | documented degenerate: {'gpt2'}

ranked frac_cnn (14s)

ranked frac_vit (28s)

ranked rand_cnn (55s)
 ranked rand_vit (68s)
 ranked nat_cnn (83s)
 ranked nat_vit (96s)
 ranked pixels (112s)
 ranked frac_cnn_inorm (126s)
 ranked frac_vit_cls (140s)

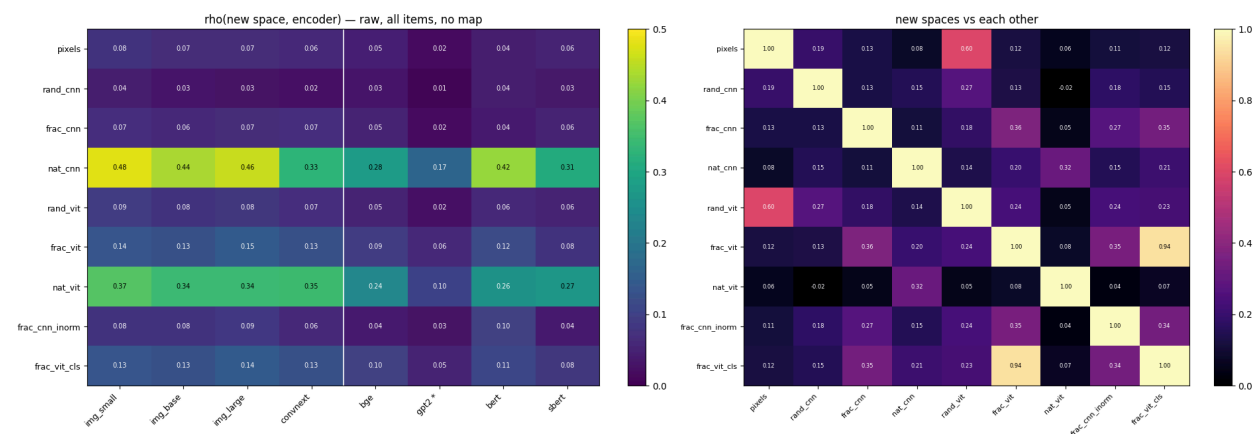
rho(new space, encoder) - all items, no map

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.079	0.067	0.067	0.063	0.047	0.018	0.043	0.056
rand_cnn	0.042	0.032	0.028	0.025	0.032	0.007	0.041	0.035
frac_cnn	0.071	0.064	0.066	0.069	0.052	0.019	0.037	0.056
nat_cnn	0.477	0.436	0.459	0.328	0.282	0.168	0.424	0.312
rand_vit	0.089	0.078	0.081	0.069	0.052	0.024	0.062	0.056
frac_vit	0.138	0.130	0.148	0.126	0.090	0.059	0.116	0.075
nat_vit	0.369	0.342	0.342	0.347	0.236	0.104	0.261	0.272
frac_cnn_inorm	0.080	0.076	0.086	0.064	0.040	0.029	0.096	0.040
frac_vit_cls	0.135	0.128	0.142	0.127	0.096	0.052	0.106	0.082

ConvNeXt row measured in this run (the reproduction gate compares it with Table C12m):
 {'img_small': 0.397, 'img_base': 0.392, 'img_large': 0.391, 'bge': 0.302, 'gpt2': 0.166, 'bert': 0.305, 'sbert': 0.33}

9

G9 raw shape agreement (* GPT-2 collapsed, stratified out of means)



, 5/30 subsamples (39s)

10/30 subsamples (79s)

15/30 subsamples (118s)

20/30 subsamples (157s)

25/30 subsamples (196s)

30/30 subsamples (235s)

subsample SD of rho:

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.0032	0.0031	0.0036	0.0025	0.0038	0.0019	0.0023	0.0040
rand_cnn	0.0054	0.0070	0.0080	0.0043	0.0102	0.0081	0.0113	0.0069
frac_cnn	0.0052	0.0049	0.0061	0.0045	0.0084	0.0090	0.0094	0.0063
nat_cnn	0.0053	0.0057	0.0061	0.0055	0.0096	0.0078	0.0090	0.0079
rand_vit	0.0041	0.0039	0.0048	0.0029	0.0052	0.0047	0.0057	0.0050
frac_vit	0.0065	0.0066	0.0081	0.0058	0.0106	0.0089	0.0121	0.0081
nat_vit	0.0056	0.0057	0.0068	0.0039	0.0056	0.0053	0.0066	0.0055

shuffle null (mean): {'frac_cnn', 'img_base': np.float64(-0.0003), ('frac_cnn', 'bge'): np.float64(-0.0007), ('frac_vit', 'img_base'): np.float64(0.0005), ('frac_vit', 'bge'): np.float64(0.0003)}, Reality fraction R (0 = architecture only, 1 = as good as ImageNet); NaN = bracket < 0.02

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
cnn	0.066	0.077	0.088	0.147	0.080	0.074	-0.011	0.075
vit	0.176	0.198	0.256	0.207	0.203	0.439	0.271	0.090

PRE-REGISTERED VERDICTS

P1 [cnn] CONFIRMED 8/8

P1 [vit] CONFIRMED 8/8

P2 [cnn] CONFIRMED 4/4

P2 [vit] CONFIRMED 4/4

P3 [cnn] INCONCLUSIVE R_img - R_txt 95% interval [-0.106, +0.100]

P3 [vit] INCONCLUSIVE R_img - R_txt 95% interval [-0.053, +0.135]

P5 [cnn] FALSIFIED 0/4

P5 [vit] CONFIRMED 4/4

R [cnn]: image mean 0.094, healthy-text mean 0.048, GPT-2 0.074 (stratified out)

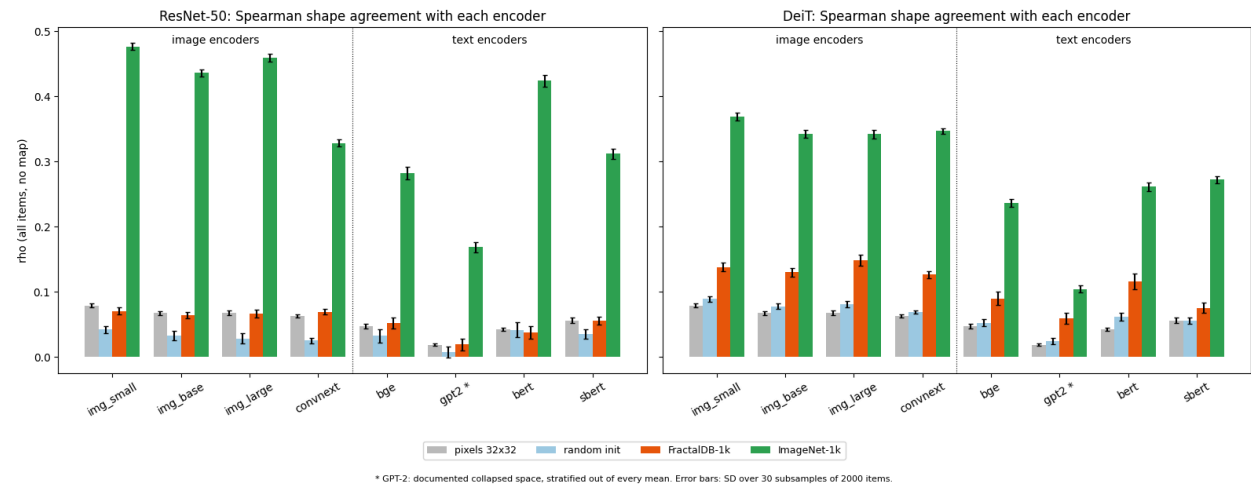
R [vit]: image mean 0.209, healthy-text mean 0.188, GPT-2 0.439 (stratified out)

SENSITIVITY (max |delta rho| over the 8 encoders; flag > 0.02):

ViT cls-only vs cls+patch 0.0103 ok

frac_cnn: ImageNet vs repo normalization 0.0588 FLAG

9



9

step	step_name	id	measure	experiment	result	control_or_chance	what_it_proves	new	source
S0	Instrument validity	0.1	Row alignment by shuffle test	All caches, 9,533 COCO train2017 items; ridge or rank correlation intact vs row-shuffled. Two id namespaces (request position vs COCO id) never joined by id.	Honest fits beat shuffled by 0.545-0.863 (threshold 0.2); scale-extension encoders 0.26-0.85	Shuffled = 0.00	Row i is the same item in every cache - a precondition, not PRH evidence	Project method	E.15; H1
S0	Instrument validity	0.2	Artifact certifica	47 automated checks over	No hard failures; 2 flags, both the	-	Caches are well-formed and aligned	Project method	D.5; H1

			tion (H1)	every cache: shapes, norms, alignment, rows/dim	documented GPT-2 collapse				
S0	Instrument validity	0.3	Statistical power: rows per input dimension	Every fit checked against a pre-registered floor of 5 rows/dim. Exp A run 1 (1,646 samples) vs run 2 (10,000), identical pipeline	All fits 5.9-9.8 rows/dim; Exp A run 1 FAILED on sample starvation, run 2 PASSED	Floor = 5	Failures are diagnosable as power, not as absence of convergence	Methodological contribution (report s.5)	s.3.2-3.3; H1
S0	Instrument validity	0.4	Null calibration by permutation (G5)	Permutation nulls for k-NN overlap and CKA across 21 encoder pairs	k-NN null = k/N exactly; CKA null = +0.094 and rises with width. Correcting it widens objective-over-lineage gap +0.024 -> +0.181	Permutation null	Uncorrected CKA manufactures agreement that favoured the lineage pair	Known bias; its effect on this project's conclusion is project-specific	G5; C.13.9
S1	'Well-trained' - learned, not architectural	1.1	Linear CKA, trained vs random weights (Exp A)	GPT-2 vs Pythia-160M (deliberately hostile pair), WikiText-103 10k passages, layerwise CKA	Diagonal mean 0.424 (pass >0.35); trained/random 6.3x (pass >=3x)	Random-weights control 0.067 (pass <0.15)	Similarity is produced by training, not by architecture	Replication (Kornblith 2019) on a hostile pair	s.3.4; C.7

S1	'Well-trained' - learned, not architectural	1.2	Stitching R2 and depth signature (Exp A)	Same pair; one linear map per layer, held-out R2	Peak L11 0.797 (pass >0.70); trained curve rises 0.59 -> 0.80 with depth	Random 0.173 at L11 and DECAYS 0.38 -> 0.09 - opposite slope, 8x gap	Convergence builds with depth - learned, not inherited	Replication (Bansal 2021)	s.3.4
S1b	'Well-trained' ON REALITY - is it the data?	F.1	Shape agreement: Fractal DB ResNet -50 vs each of the 8 encoders	ResNet-50 trained ONLY on FractalDB -1k (1M rendered fractals, 1k formula-defined classes, zero natural images); 2048-d global-average-pooled features of the same 9,533 COCO images; Spearman of pairwise cosine, all items, no map	FRACTAL: img_small 0.071, img_base 0.064, img_large 0.066, convnext 0.069, bge 0.052, gpt2 0.019, bert 0.037, sbert 0.056. ImageNet twin: img_small 0.477, img_base 0.436, img_large 0.459, convnext 0.328, bge 0.282, gpt2 0.168, bert 0.424, sbert 0.312. P1 CONFIRMED (8/8 encoders below the ImageNet twin by > 2 SD)	Random twin: img_small 0.042, img_base 0.032, img_large 0.028, convnext 0.025, bge 0.032, gpt2 0.007, bert 0.041, sbert 0.035. Shuffle null mean -0.0007	Fractal training stops short of what natural images give on nearly every encoder: the convergence is data-driven, as PRH's mechanism says	Candidate new - no prior test of FDSL models against a cross-modal PRH roster found in searches	G9 (this notebook)

S1 b	'Well-trained' ON REALITY - is it the data?	F . 2	Shape agreement: Fractal DB DeiT vs each of the 8 encoders	DeiT (ViT) trained ONLY on FractalDB -1k; cls+patch pooling to match the DINOv2 caches; same items and statistic	FRACTAL: img_small 0.138, img_base 0.130, img_large 0.148, convnext 0.126, bge 0.090, gpt2 0.059, bert 0.116, sbert 0.075. ImageNet twin: img_small 0.369, img_base 0.342, img_large 0.342, convnext 0.347, bge 0.236, gpt2 0.104, bert 0.261, sbert 0.272. P1 CONFIRMED (8/8 encoders below the ImageNet twin by > 2 SD)	Random twin: img_small 0.089, img_base 0.078, img_large 0.081, convnext 0.069, bge 0.052, gpt2 0.024, bert 0.062, sbert 0.056. Shuffle null mean +0.0005	Same P1 verdict on a transformer - not a CNN idiosyncrasy	Candidate new (as F.1)	G9
S1 b	'Well-trained' ON REALITY - is it the data?	F . 3	Brackets: random-init and ImageNet twins, pixel floor	Same two architectures with random weights (lower bracket) and trained on ImageNet -1k (upper bracket); 32x32 raw pixels	Pixels (image-encoder mean) 0.069; random CNN/ViT 0.032/0.079; fractal 0.067/0.136; ImageNet 0.425/0.350. P2 cnn CONFIRMED (4/4),	Brackets ARE the controls	Places the fractal result on a scale from 'architecture only' to 'architecture + reality'	Project design	G9

				as the floor	vit CONFIRMED (4/4); P5 cnn FALSIFIED, vit CONFIRMED				
S1b	'Well-trained' ON REALITY - is it the data?	F . 4	Reality fraction R, image vs text encoders	R = (rho_fractional - rho_random) / (rho_natural - rho_random) per encoder; undefined if the bracket < 0.02	[cnn] R image 0.094, healthy text 0.048, R_img-R_text 95% [-0.106, +0.100] -> P3 INCONCLUSIVE; GPT-2 R 0.074 (stratified out); [vit] R image 0.209, healthy text 0.188, R_img-R_text 95% [-0.053, +0.135] -> P3 INCONCLUSIVE; GPT-2 R 0.439 (stratified out)	Subsample interval (30 x 2,000 items); R undefined where the bracket < 0.02	No reliable image-vs-text difference in R: the language-shared structure is not specially dependent on natural-image training at this scale (or the brackets are too narrow to tell)	Candidate new metric	G9
S1b	'Well-trained' ON REALITY - is it the data?	F . 5	Sensitivity: pooling and normalization	DeiT cls-only vs cls+patch; repo normalization vs ImageNet normalization	ViT cls-only vs cls+patch: max d-rho 0.0103; frac_cnn: ImageNet vs repo normalization: max d-rho	-	Checks the result is not a preprocessing artifact (the giant cls vs cls+patch lesson, E.16)	Project practice	G9

					0.0588 FLAG				
S1 b	'Well-trained' ON REALITY - is it the data?	F .6	Whiteni ng rescue of the collaps ed fractal CNN (C.11 move)	frac_cnn / rand_cnn are collapsed (mean pair-cos 0.85 / 1.00); PCA-whit en each bracket space and re-measur e the same Spearman rho. Row correspon dence is known, so this is the READING regime (C.11), not the FINDING regime (C.12). Secondar y diagnostic ; the pre-regist ered raw verdicts stand.	PENDING - run G9 cells 17-18. Raw P5-cnn FALSIFIED (frac_cnn at the pixel floor) may flip to CONFIRM ED once collapse is whitened away, if the masked signal is real	Random-init twin and 32x32 pixels as floors, both whitened the same way; row-shuffle null	If whitening flips P5-cnn to CONFIRMED: the raw CNN 'failure' was collapse, and the fractal CNN does carry a small real slice of shared structure (size = R_white, typically << natural images). If it stays FALSIFIED: the fractal CNN genuinely shares nothing at this pooling	Third instance of the E.6/E.12 lesson (after GPT-2 and ConvNeXt): raw agreeme nt on a degener ate space measures the post-coll apse residual, not what is recovera ble	G9
S2	'Independent' - no shared ancestry or recipe	2 .1	Cross-li neage transfer (G4)	SigLIP 2 (independ ent lineage, no DINOv2 ancestry) given one map into the frozen	94.2% of a natively-fitt ed head - inside the 93.8-96.5 % within-famil y band	Random-map control at chance	Shared space is not a family trait	Extensio n (cross-lin eage, not simultan eously cross-obj ective;	C.13; G4; G0-exact

				4-space hub; frozen caption head				still a ViT)	
S2	'Independent' - no shared ancestry or recipe	2 . 2	Cross-architecture transfer (E.2)	ConvNeXt-base (convnet, ImageNet-22k supervised) held out of the hub, one entry map	96.7% of native - ABOVE the band and above SigLIP 2 in the same run; 99.3% in the 7-space hub (G10)	Control 0.001 = chance	Architecture does not bound the claim (objective still confounded)	New in this project	E.2; G10
S2	'Independent' - no shared ancestry or recipe	2 . 3	Supervision regime (E.10)	Label-free (DINOv2, GPT-2, BERT), pair-supervised (bge, SigLIP) and 22k-class supervised (ConvNeXt) encoders in one hub	All carry a head at 93-97% of native	-	Convergence does not require a shared supervision signal - keeps the thesis non-circular	New in this project	E.10
S2	'Independent' - no shared ancestry or recipe	2 . 4	What explains pairwise agreement (G5, exploratory)	21 pairs grouped by shared objective / lineage / modality	Objective +53x > lineage +44x > modality +29x; top pair bge-SBERT shares objective, not lineage	Permutation-calibrated	Supervision modulates HOW MUCH encoders agree, not WHETHER	New in this project (exploratory)	G5; E.10

S3	'Up to a linear transformation'	3.1	MLP vs ridge, controlled (F1/F2)	Distillation corpus 38k rows (74.2 rows/dim), MLP vs closed-form ridge, two independent runs	Held-out cosine 0.9183 = 0.9183, gain +0.0000, both runs	Same data, same split	Convergence is linear - capacity beyond a matrix adds nothing	New in this project (controlled test)	C.9; F1/F2
S3	'Up to a linear transformation'	3.2	Adapter: linear vs nonlinear (Exp B)	MobileCLIP-S1 -> SigLIP 2, 4,000 COCO val2017, 3,000/1,000 split	Ridge cosine 0.900; MLP 0.903 (+0.003)	-	Deployed relationship is linear	Consistent with stitching literature	s.4; B2
S3	'Up to a linear transformation'	3.3	Related, not rigid: rotation-only vs linear	Orthogonal Procrustes vs ridge: Exp B pair; 21 hub pairs (G6); raw-space similarity transform (K1)	Procrustes R2 0.038 vs ridge 0.592 (Exp B), 69x singular-value spread; 0.194 vs 0.667 over 21 pairs; similarity transform 0.281 vs 0.667	-	The strong (rotation-only) sub-claim FAILS: spaces are linearly related but anisotropic	New in this project	s.4; G6; K1
S4	Across modalities - the 'Platonic' claim proper	4.1	Image-only -> text-only retrieval (C.8)	DINOv2-base (never saw text) -> bge-m3 (never saw an image); 8,000 COCO train2017 pairs, 9.1	R2 0.511; R@1 0.358 = 40.9% of a matched SigLIP ceiling (0.875); pre-registered verdict: measurable but weak	358x chance; raw cross-space floor 0.001; shuffle gap 0.641	Cross-modal structure exists without joint training	Replication of PRH's cross-modal claim at small scale	C.8

				rows/dim, ridge					
S4	Across modalities - the 'Platonic' claim proper	4 . 2	Raw cross-modal shape agreement	Spearman of pairwise cosine distances, 8 encoders, all 9,533 items, no map	Image-text mean rho 0.290 (8 encoders); top pair bge-SBERT 0.768; contrastive text pairs reach 0.76-0.86 in the scale extension	rho ~ 0 unrelated	Cross-modal agreement is real but modest; same-objective text agrees far more	Replication (PRH-style measurement)	C.13.9; E.12; E.16
S4	Across modalities - the 'Platonic' claim proper	4 . 3	Shared subspace size (G6)	CCA across pairs; intrinsic-dimension estimates	Cross-modal pairs share as few as 6 of 64 CCA directions; intrinsic dimension 11.2-19.9 inside 768-2048-d spaces (<3%)	-	What is shared is a small, low-dimensional structure	New in this project	G6
S4	Across modalities - the 'Platonic' claim proper	4 . 4	Reach into the shared hub (E.6)	Each encoder's held-out R2 for its own map into the 7-space hub	Image encoders 0.46-0.57; all four text spaces 0.02-0.05, uniformly	-	A whitened hub is largely an image-side object; text touches few of its directions	New in this project	E.6
S5	'As models grow stronger' - scaling	5 . 1	Capacity sweep (C.10)	DINOv2 small/bas e/large -> bge, pooling and rows/dim	40.9 / 53.5 / 58.3% of ceiling; +15.3 points per decade of parameter	Matched ceiling	Alignment rises with capacity in this range	Replication of PRH scaling	C.10

				(11.1) held constant	s; fit R2 0.949; six monotone measures			at small scale	
S5	'As models grow stronger' - scaling	5 . 2	Pre-registered scale extension (E.16)	DINOv2-giant (cls+patch) and Qwen-4B added to matched ladders; thresholds fixed before running	Giant rho 0.339 < 0.418 FALSIFIED ; Qwen-4B rho 0.307 < 0.320 FALSIFIED ; both ladders rise then fall	Pooling artifact ruled out (cls-only 0.238 discarded)	Naive 'more capacity -> more alignment' is NOT supported at this scale	New in this project (pre-registered negative)	E.16
S5	'As models grow stronger' - scaling	5 . 3	Raw agreement vs transfer under scaling (E.16 P4)	Same ladders, both quantities	delta-rho -0.079 while delta-transfer +7.1 points - opposite directions	-	Scaling moves the two quantities apart	New in this project	E.16
S6	Mechanism - what drives convergence	6 . 1	Isotropy rescue (C.11)	GPT-2 caption space (effective rank 6.5) whitened, label-free, fitted on training rows only	R@1 0.130 -> 0.478 = 102.6% of a 1.2B-pair contrastive encoder (101.4% at a second scale); eff. rank 6.5 -> 490.8; iso-retrieval corr +0.94	Same protocol, unwhitened	Isotropy, not similarity training, is what a space needs to be alignable	New in this project	C.11; E1.2
S6	Mechanism - what drives	6 . 2	Encoder ladder as a predictor	GPT-2 / BERT / SBERT as text side, same	Eff. rank 6.5 / 65.0 / 93.4 -> R@1 0.160 / 0.362 /	-	The mechanism predicts before measuring; isotropy	New in this project	C.14

	convergence		on (C.14)	image encoder, pre-registered	0.510; rank corr +1.00 (P1 confirmed); P2 partial		necessary, not sufficient		
S6	Mechanism - what drives convergence	6.3	Meta-finding: read vs find	C.11 (known correspondence) vs C.12 (unknown correspondence)	Whitening helps reading (+102.6%) and destroys finding (39.7% -> 0.1%)	-	One geometric property, opposite prescriptions - two instances (corrected down from three)	New in this project	C.11; C.12; E.8
S7	Self-identifying correspondence	7.1	Pair-free recovery by Gromov-Wasserstein (C.12)	Two contrastive image-text encoders, zero paired examples	39.7% exact matches; map fitted from that assignment R@1 0.252 (40% of ceiling), AUC 0.906	595x chance (0.067%)	The correspondence is self-identifying from geometry alone - bounded: weakest-independence pair	Replication in spirit (vec2vec 2025) by a different method	C.12
S8	One shared space, used	8.1	Zero-shot hub transfer (C.13)	5 spaces over 9,533 images, one linear map each into a 512-d whitened hub; caption head trained on img_small only	img_base 95.9%, img_large 92.9% of native - absolute R@1 RISES on unseen encoders	Random-map control 0.000 (chance 0.001)	The shared space exists by use, not only by correlation	New in this project	C.13

S8	One shared space, used	8 . 2	Composition and compatibility (G1, G3)	7-space hub; all 42 ordered pairs vs bespoke hubs	95.6% retention over 36/41 pairs (exceptions all write INTO GPT-2); 42/42 pairs shared, mean 103.6%; GPT-2 source 95.1% vs target 24.8%	Control 0.004 vs chance 0.001	One hub serves all members; collapsed spaces are readable out of, not writable into	New in this project	C.13; G1; G3
S8	One shared space, used	8 . 3	Deployed adapter (Exp B)	MobileCLIP-S1 -> SigLIP 2 ridge matrix, shared Qdrant index, Core ML fp16	R@1/5/10 = 88.4 / 95.3 / 97.2% of ceiling; verification AUC 0.999; hubness skew 4.28 -> 1.61; 0.79 MB	Raw cross-space baseline = exact chance	PRH is load-bearing in a working two-tier system	Uncommon application (report s.5)	s.4; D.2; D.3
S9	Boundaries and falsifications	9 . 1	Local, not global (G5)	CKNNA (the literature's metric) vs the project's own k-NN measure, 21 pairs	21/21 pairs higher locally than at the global limit; the two metrics agree at Spearman +0.81	k-NN null = k/N	Agreement is concentrated in local neighbourhoods	Replication of CKNNA + project finding	G5; C.13.9
S9	Boundaries and falsifications	9 . 2	Raw geometry does not predict	ConvNeXt's raw rho vs its hub transfer; second	Lowest raw rho of any image encoder (0.391-0.397 to DINOv2,	-	Raw shape agreement bounds what is VISIBLE without fitting, not what is	New in this project	E.12; E.6

			transfer (E.12)	instance of E.6	0.166-0.330 to text) yet best transfer 96.7%		RECOVERABLE		
S9	Boundaries and falsifications	9.3	Width cliff explained (E.8)	Hub-width sweep; retained-rank sweep in steps of 8	Step at the source's ambient dimension, 6/6 curves; sigma 0.0054 vs 10.79; holds to k=760 then 0.009; repaired by truncation (0.878) or head alpha 1e3 (0.917)	Three off-dimension widths: nothing	A numerical artifact of least squares, NOT a property of shared representations	New in this project	E.3; E.8; G11-G16
S9	Boundaries and falsifications	9.4	Falsification ledger	Every proposed explanation tested and recorded	Nine findings; eleven explanations falsified, two of them the project's own	-	Claims that survived did so against recorded alternatives	Project practice	s.7.4; Results_Summary
S9	Boundaries and falsifications	9.5	Scope boundary	Roster width vs PRH's evidence base	Encoders here are 768-2,048-d; PRH's evidence extends to 70B-parameter LLMs	-	Results bound the claim at this scale; they do not test PRH's large-model regime	Boundary	E.16; viva brief

written:

G9_ablation_views.png

G9_fractal_brackets.png

G9_results.json

G9_rho_heatmaps.png

G9_whiten_dim_sweep.csv

G9_whiten_dim_sweep.png

G9_whitening_centered.csv

G9_whitening_drop1.csv

G9_whitening_heatmap.png

G9_whitening_pca.csv

G9_whitening_raw.csv

G9_whitening_rescue.json

G9_whitening_rescue.png

G9_whitening_white.csv

PRH_Evidence_Table.csv

PRH_Evidence_Table.md

PRH_Evidence_Table.xlsx, fit transforms on 8533 train rows, measure rho on 1000 held-out rows

held-out geometry per view (mean pair-cosine):

	space	raw_cos	centered_cos	pca_cos	white_cos
pixels	-0.000	0.000	0.000	0.000	
rand_cnn	0.998	-0.000	-0.000	0.000	
frac_cnn	0.847	0.004	0.004	0.001	
nat_cnn	0.569	0.000	0.000	0.000	
rand_vit	0.259	0.005	0.005	0.000	
frac_vit	0.371	0.003	0.004	0.000	
nat_vit	0.125	0.000	0.000	0.000	

,RAW rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.076	0.064	0.060	0.062	0.044	0.015	0.042	0.051
rand_cnn	0.040	0.027	0.025	0.016	0.044	-0.009	0.033	0.039
frac_cnn	0.070	0.062	0.068	0.066	0.068	-0.013	0.027	0.061
nat_cnn	0.479	0.435	0.464	0.322	0.292	0.182	0.422	0.316
rand_vit	0.092	0.082	0.085	0.068	0.058	0.019	0.074	0.053
frac_vit	0.144	0.140	0.155	0.121	0.107	0.064	0.133	0.085
nat_vit	0.367	0.342	0.347	0.348	0.231	0.113	0.258	0.262

CENTERED rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.076	0.064	0.060	0.062	0.044	0.014	0.042	0.051
rand_cnn	0.054	0.041	0.040	0.049	0.025	0.010	0.021	0.020
frac_cnn	0.072	0.064	0.072	0.081	0.060	0.022	0.055	0.066
nat_cnn	0.470	0.425	0.446	0.382	0.356	0.174	0.365	0.415
rand_vit	0.083	0.073	0.073	0.068	0.049	0.015	0.051	0.050
frac_vit	0.125	0.122	0.137	0.120	0.086	0.053	0.131	0.096
nat_vit	0.378	0.334	0.330	0.339	0.258	0.122	0.258	0.300

DROP1 rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.053	0.050	0.047	0.052	0.041	0.018	0.048	0.048

rand_cnn	0.057	0.047	0.045	0.053	0.033	0.011	0.035	0.039
frac_cnn	0.066	0.055	0.056	0.061	0.041	0.014	0.033	0.045
nat_cnn	0.346	0.294	0.269	0.332	0.215	0.121	0.182	0.232
rand_vit	0.095	0.096	0.106	0.095	0.076	0.039	0.103	0.083
frac_vit	0.072	0.060	0.059	0.076	0.036	0.024	0.039	0.039
nat_vit	0.286	0.241	0.214	0.286	0.164	0.092	0.150	0.175

PCA rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.076	0.064	0.060	0.062	0.044	0.014	0.042	0.050
rand_cnn	0.054	0.041	0.040	0.049	0.026	0.010	0.021	0.020
frac_cnn	0.072	0.065	0.072	0.081	0.060	0.022	0.055	0.066
nat_cnn	0.470	0.425	0.446	0.382	0.355	0.174	0.364	0.415
rand_vit	0.083	0.073	0.073	0.068	0.049	0.015	0.051	0.050
frac_vit	0.125	0.122	0.137	0.120	0.086	0.053	0.131	0.096
nat_vit	0.379	0.334	0.330	0.339	0.258	0.122	0.258	0.300

WHITE rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.025	0.024	0.023	0.024	0.016	0.007	0.017	0.017
rand_cnn	0.031	0.028	0.026	0.030	0.019	0.007	0.017	0.020
frac_cnn	0.042	0.037	0.037	0.042	0.022	0.015	0.028	0.026
nat_cnn	0.153	0.119	0.098	0.149	0.069	0.039	0.064	0.075
rand_vit	0.031	0.026	0.025	0.030	0.019	0.011	0.017	0.019

frac_vit	0.035	0.031	0.030	0.036	0.021	0.013	0.020	0.023
nat_vit	0.143	0.115	0.096	0.136	0.069	0.039	0.066	0.074

,Which operation moves the ordering? frac_cnn image-encoder mean per view (held-out):

view	frac_cnn	pixels	random	ImageNet	frac>pixels & frac>random?
raw	0.067	0.066	0.027	0.425	YES
centered	0.072	0.066	0.046	0.431	YES
drop1	0.060	0.051	0.051	0.310	YES
pca	0.072	0.065	0.046	0.431	YES
white	0.040	0.024	0.029	0.130	YES

frac_vit image-encoder mean per view (held-out):

raw	frac 0.140 (pixels 0.066, random 0.082, ImageNet 0.351)
centered	frac 0.126 (pixels 0.066, random 0.074, ImageNet 0.345)
drop1	frac 0.067 (pixels 0.051, random 0.098, ImageNet 0.257)
pca	frac 0.126 (pixels 0.065, random 0.074, ImageNet 0.345)
white	frac 0.033 (pixels 0.024, random 0.028, ImageNet 0.122)

Reading: absolute rho FALLS under whitening; the point is the ORDERING (frac above whitened pixels/random), and WHICH view first makes it hold. If 'centered' already orders frac>pixels, the effect is anisotropy in the mean; if only 'white' does, it needs full variance equalization.

, held-out whitened subsamples 10/30

held-out whitened subsamples 20/30

held-out whitened subsamples 30/30

HELD-OUT WHITENED DIAGNOSTIC verdicts (NOT the pre-registered raw test - a separate reading question)

P1w [cnn] CONFIRMED 8/8

P1w [vit] CONFIRMED 8/8

P2w [cnn] CONFIRMED 4/4

P2w [vit] CONFIRMED 4/4

P3w [cnn] INCONCLUSIVE R_img-R_txt 95% [-0.066, +0.283]

P3w [vit] INCONCLUSIVE R_img-R_txt 95% [-0.165, +0.121]

P5w [cnn] CONFIRMED 4/4

P5w [vit] CONFIRMED 4/4

R_white [cnn]: image 0.112, healthy-text 0.140

R_white [vit]: image 0.056, healthy-text 0.056

RESULT [cnn]: a small residual signal became visible after de-anisotropization (11% of the natural-image bracket, held-out); P2w CONFIRMED, P5w CONFIRMED

RESULT [vit]: a small residual signal became visible after de-anisotropization (6% of the natural-image bracket, held-out); P2w CONFIRMED, P5w CONFIRMED

-> whitening EXPOSES the ordering (small residual > controls), it does NOT raise absolute rho, and it is NOT the Atlas sense of recoverable (fitted entry-map + frozen head) -- that is future work.

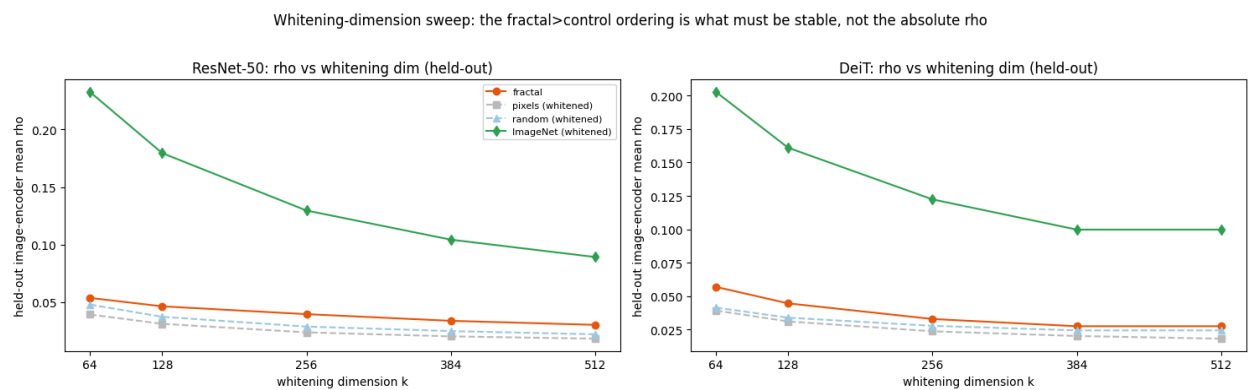
	, k	arch	eff_k	frac	pixels	random	imagenet	frac_gt_pixels	frac_gt_random	ordering_holds
64	cnn	64	0.0537	0.0393	0.0479	0.2326	True	True	True	
64	vit	64	0.0570	0.0393	0.0415	0.2027	True	True	True	
128	cnn	128	0.0464	0.0312	0.0373	0.1795	True	True	True	
128	vit	128	0.0447	0.0312	0.0340	0.1610	True	True	True	
256	cnn	256	0.0396	0.0238	0.0288	0.1296	True	True	True	
256	vit	256	0.0330	0.0238	0.0279	0.1224	True	True	True	

384	cnn	384	0.0338	0.0203	0.0249	0.1043	True	True	True
384	vit	382	0.0276	0.0203	0.0245	0.0998	True	True	True
512	cnn	512	0.0303	0.0183	0.0221	0.0892	True	True	True
512	vit	382	0.0276	0.0183	0.0245	0.0998	True	True	True

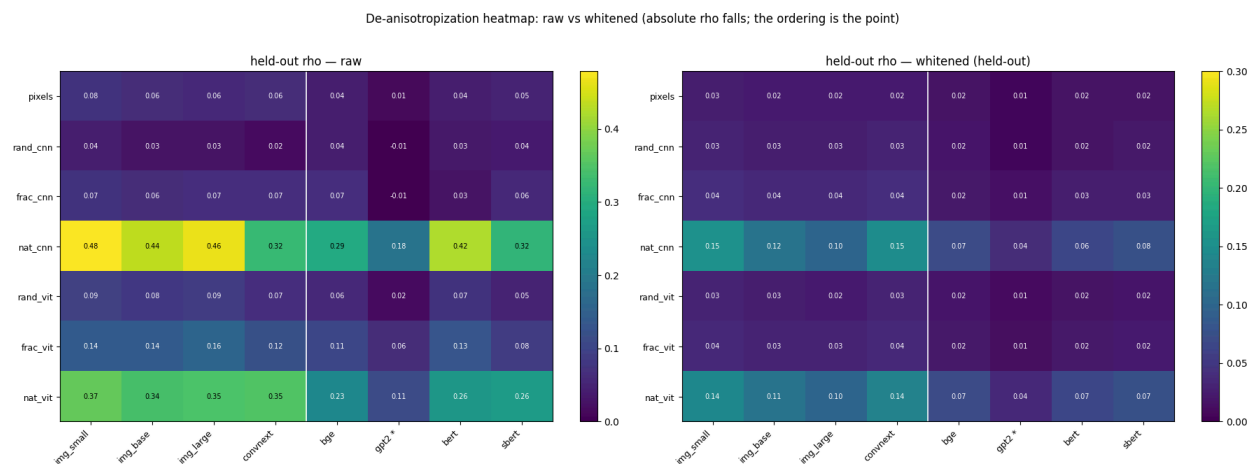
ordering (fractal > whitened pixels AND > whitened random) holds across ALL k:

ResNet-50: YES - conclusion robust to k

DeiT: YES - conclusion robust to k

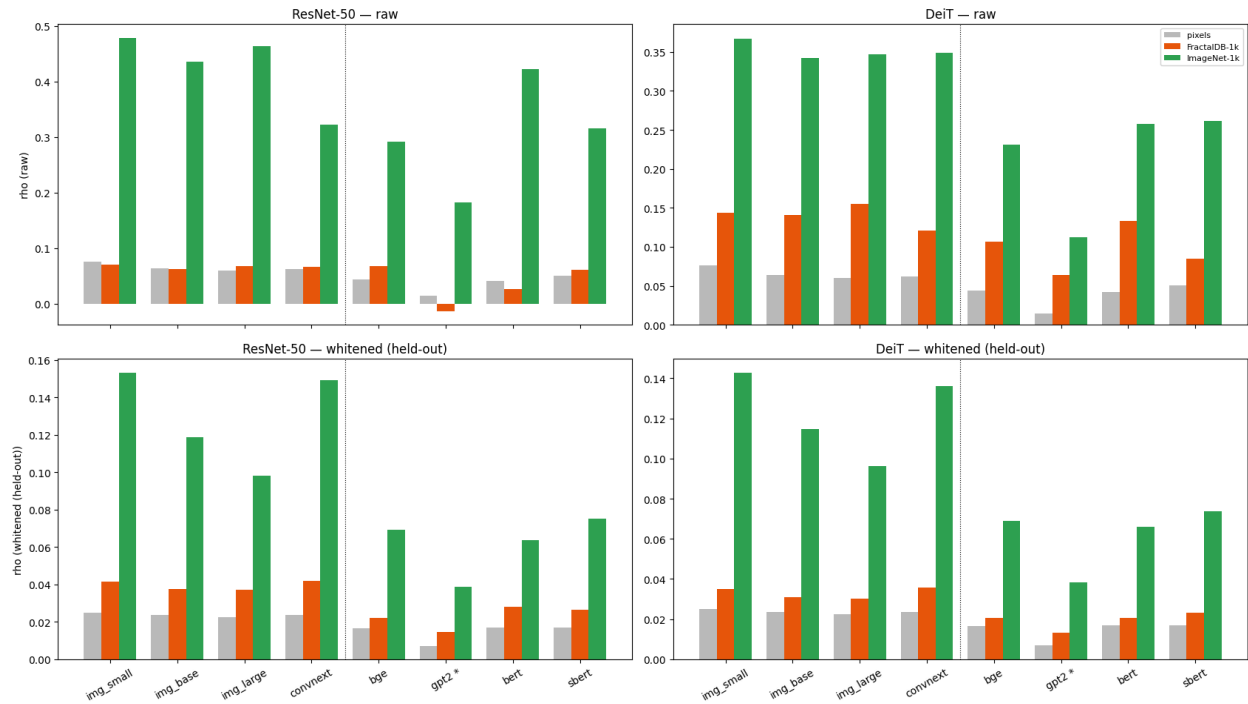


,

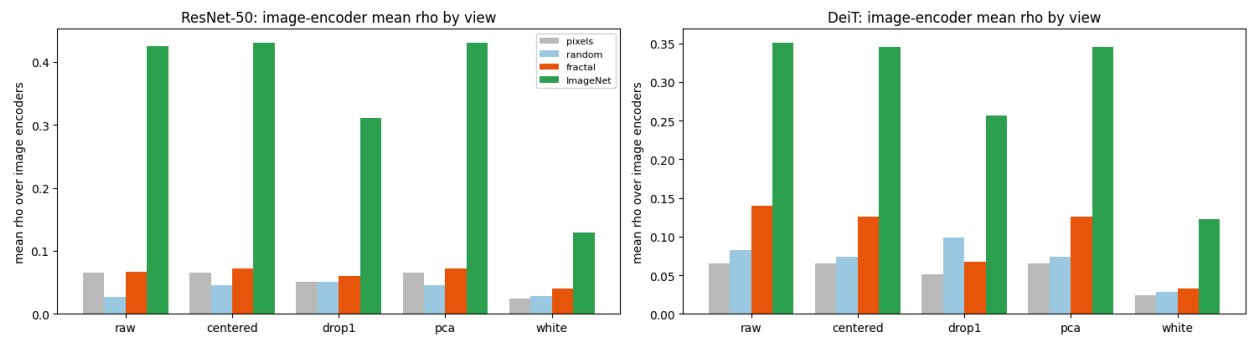


9

De-anisotropization (held-out): a small residual fractal signal above whitened controls; natural images remain far stronger (* GPT-2 collapsed, stratified out)



Which operation makes the fractal ordering visible? (held-out)



evidence table rewritten with F.6 filled -> PRH_Evidence_Table.xlsx

written:

G9_whitening_centered.csv

G9_whitening_drop1.csv

G9_whitening_heatmap.png

G9_whitening_pca.csv

G9_whitening_raw.csv

G9_whitening_rescue.json

G9_whitening_rescue.png

G9_whitening_white.csv

Read-out: if the CNN is rescued by whitening, the raw P5-cnn 'failure' was collapse masking recoverable structure (E.6/E.12 lesson), not the absence of it. If it is NOT rescued, the fractal CNN genuinely carries no image-encoder-shared structure at this pooling. Report whichever the numbers show.

G10:

```
Mounted at /content/drive
```

```
SYNTHETIC=False
```

```
DATA_DIR=/content/drive/MyDrive/convergence_experiment exists=True
```

```
G9_OUT=/content/drive/MyDrive/convergence_experiment/G9_fractal  
exists=True
```

```
OUT=/content/drive/MyDrive/convergence_experiment/G10_fractal_hub,
```

```
PASS hub dim <= 512 and <= train rank
```

```
PASS entry-mapped hub coords have positive variance (train) mean diag  
0.649
```

```
transfer R@1: small 0.923 base 0.974 large 0.981
```

```
PASS transfer rises (or ties) with encoder quality 0.923 <=  
0.974 <= 0.981
```

```
PASS random-map control near chance ctrl R@1  
0.001
```

```
img_base: transfer 0.974 native 0.981 pct 99.3%
```

```
PASS transfer near/below native, not wildly above (img_base) pct 99.3%
```

```
PASS percent-of-native in (0,1.5] (img base)
```

```
img_large: transfer 0.981 native 0.978 pct 100.3%
```

```
PASS transfer near/below native, not wildly above (img_large) pct 100.3%
```

```
PASS percent-of-native in (0,1.5] (img_large)
```

```
PASS transfer ~invariant to source rotation 0.974 vs  
0.974
```

```
PASS transfer invariant to source global scale 0.974 vs  
0.974
```

```
PASS recall_at_k identity = 1.0
```

```
PASS recall_at_k shuffled ~ chance
```

```
ALL TESTS PASSED,
```

```
real run - skipped,
```

,8533 train / 1000 held-out rows (seed 0)

hub: 512-d whitened PCA over 5376 concatenated dims, member dims [768, 1536, 2048, 1024]

frozen caption head fitted on DINOv2-small hub coords -> (512, 1024),

	encoder	transfer_R1	native_R1	pct_native	R5	R10
img_small		0.386	0.393	0.9822	0.753	0.863
img_base		0.437	0.461	0.9479	0.795	0.889
img_large		0.453	0.485	0.9340	0.790	0.896
random-map control		0.001	NaN	NaN	0.006	0.008

Published: img_base 95.9%, img_large 92.9%, control at chance (0.001).

VERIFIED -- reproduces the hub protocol

effective transfer floor = 0.050 (max of 0.05 and 2x control 0.001)

, frac_cnn (9533, 2048)

frac_vit (9533, 384)

rand_cnn (9533, 2048)

rand_vit (9533, 384)

nat_cnn (9533, 2048)

nat_vit (9533, 384)

pixels (9533, 3072)

	transfer_R1	native_R1	pct	R5	R10	at_floor
frac_cnn	0.014	0.014	NaN	0.066	0.102	True
frac_vit	0.020	0.022	NaN	0.085	0.154	True
rand_cnn	0.012	0.012	NaN	0.053	0.093	True
rand_vit	0.017	0.018	NaN	0.065	0.102	True
nat_cnn	0.258	0.274	0.942	0.604	0.760	False
nat_vit	0.204	0.208	0.981	0.522	0.688	False
pixels	0.000	0.004	NaN	0.016	0.028	True

transfer R@1 (through frozen hub + frozen head) — the recoverable-structure number. Floor = 0.050.

[cnn] fractal 0.014 vs random 0.012, pixels 0.000, ImageNet 0.258; fractal = n/a (at floor) <-- AT FLOOR: transfer indistinguishable from useless; % of native is meaningless

[vit] fractal 0.020 vs random 0.017, pixels 0.000, ImageNet 0.204; fractal = n/a (at floor) <-- AT FLOOR: transfer indistinguishable from useless; % of native is meaningless

,PRE-REGISTERED VERDICTS

[cnn] H1 fractal>random & >pixels: FALSIFIED (vs random $d=+0.002/0.005SE$, vs pixels $d=+0.014/0.004SE$)

[cnn] H2 ImageNet>fractal: CONFIRMED ($d=+0.244/0.015SE$); fractal recoverable = n/a (at floor)

[vit] H1 fractal>random & >pixels: FALSIFIED (vs random $d=+0.004/0.006SE$, vs pixels $d=+0.021/0.004SE$)

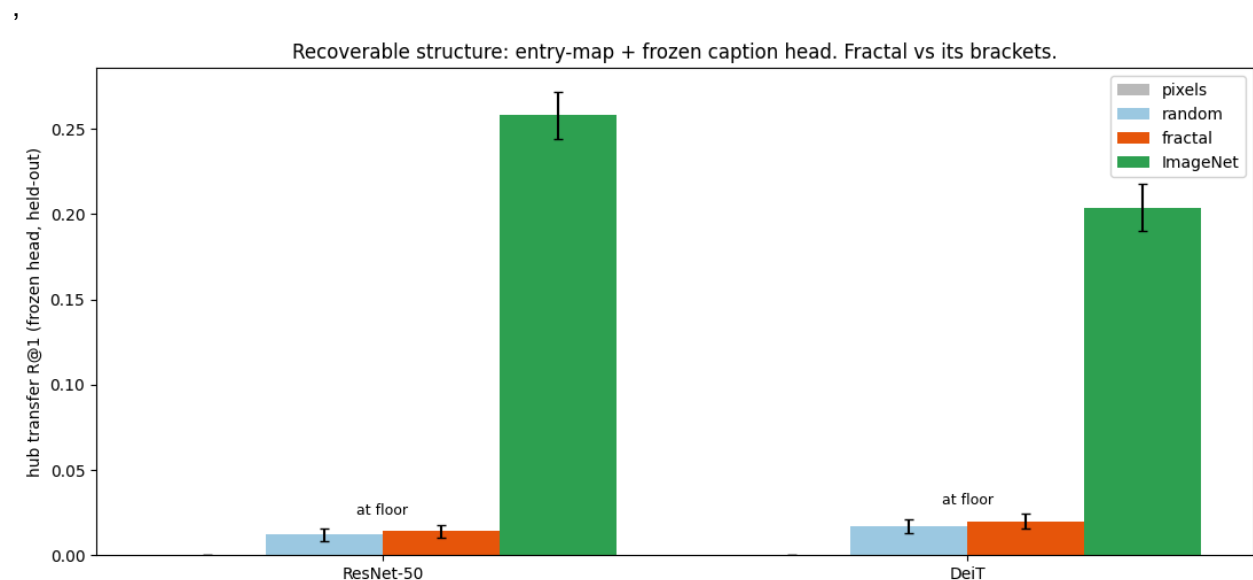
[vit] H2 ImageNet>fractal: CONFIRMED ($d=+0.184/0.015SE$); fractal recoverable = n/a (at floor)

,loaded G9 reality fractions R (visible structure)

[cnn] H3 NOT MEANINGFUL: fractal transfer 0.014 is at the floor (≤ 0.050) or does not beat random (H1 FALSIFIED). '% of native' is undefined; the fractal encoder's structure is NOT recoverable into the hub.

[vit] H3 NOT MEANINGFUL: fractal transfer 0.020 is at the floor (≤ 0.050) or does not beat random (H1 FALSIFIED). '% of native' is undefined; the fractal encoder's structure is NOT recoverable into the hub.

Reading: H3 only has meaning when fractal transfer CLEARS THE FLOOR and beats random (H1 confirmed). At the floor, '100% of native' is 100% of ~nothing -- the correct conclusion is that the fractal structure is NOT recoverable, matching H1. Natural-image encoders clear the floor and are the meaningful recoverable-structure comparison.



written:

G10_fractal_hub_transfer.png

G10_recoverable_vs_visible.png

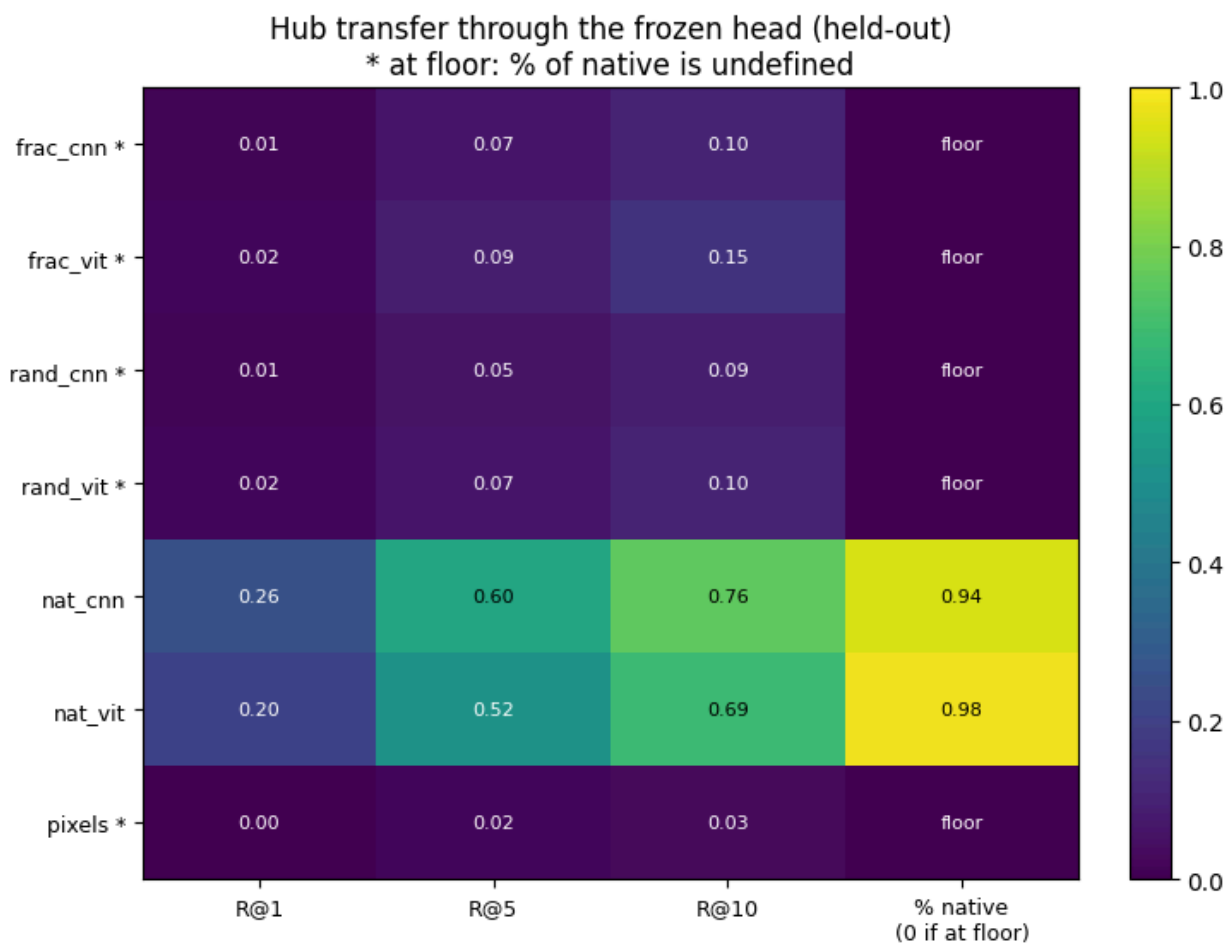
G10_results.json

G10_transfer_heatmap.png

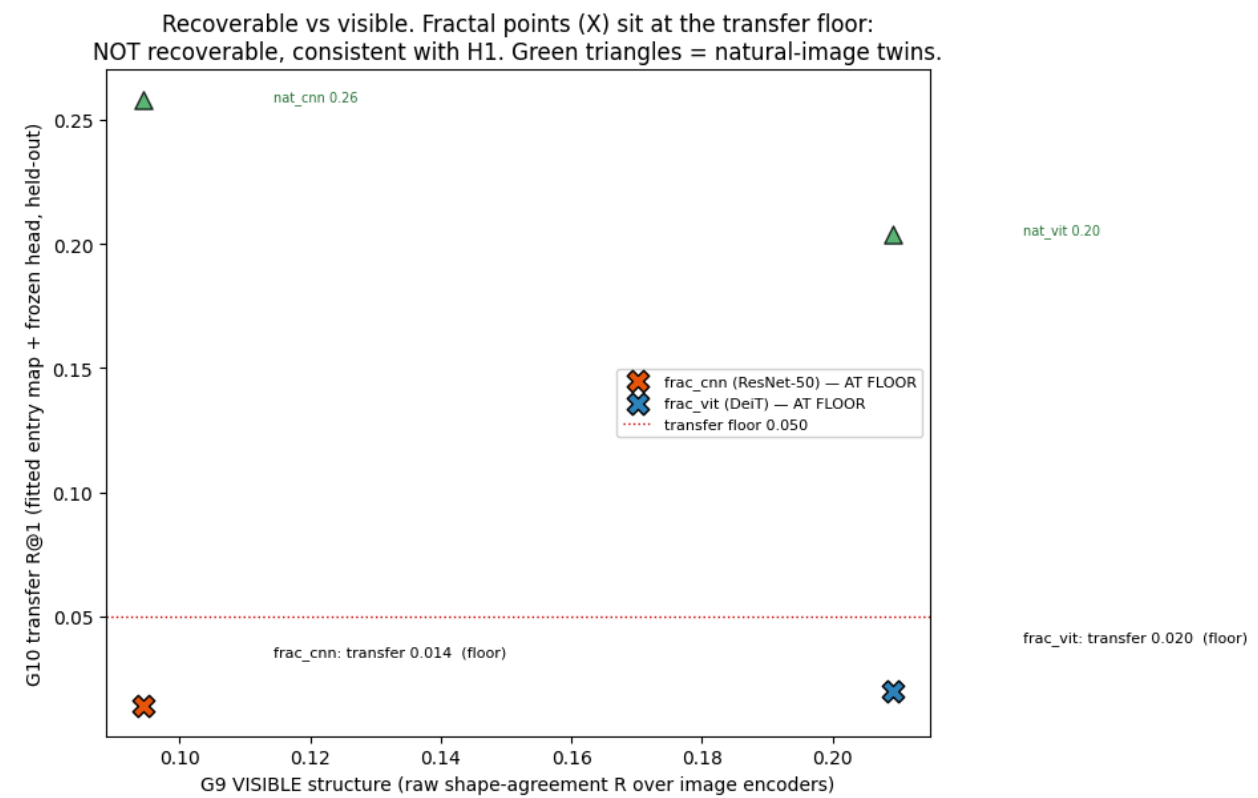
G10_transfer_table.csv

BOTTOM LINE: G9 asked if fractal structure is VISIBLE (raw geometry) -- mostly at the floor. G10 asks if it is RECOVERABLE (fitted entry map + frozen head). Answer: the fractal encoders transfer at the floor (~ 0.01 - 0.02 R@1), statistically indistinguishable from their random twins (H1 FALSIFIED). So the structure is NOT recoverable into the caption hub either -- a fitted map cannot rescue what is barely there. '% of native' reads $\sim 100\%$ only because the native head is ALSO at the floor; that ratio is meaningless and is guarded out. Natural-image encoders recover 0.20-0.26 and are the real point of comparison.

,



,



saved G10_transfer_heatmap.png, G10_recoverable_vs_visible.png

,

